British Journal of Educational Psychology (2009), 79, 695–711
© 2009 The British Psychological Society
The British Psychological Society
www.bpsjournals.co.uk

Direct and indirect effects of parents' education on reading achievement among third graders in Sweden

Eva Myrberg* and Monica Rosén
Department of Education, Gothenburg University, Gothenburg, Sweden

Background. Cultural capital in families and especially, the educational level of parents, has during the last decades been found to be the most important dimension of socio-economic influence on school performance. How the transmission of cultural capital over generations is concretized is however not yet fully investigated.

Aims. The aim is to unfold the influence of home background and more specifically, to reveal some important mediating factors between the educational levels of parents and the reading achievement levels of children.

Sample. Data comes from the Swedish participation in Progress in International Reading Literacy Study 2001 conducted by the The International Association for the Evaluation of Educational Achievement and comprises some 10,000 students in grade 3.

Methods. The effects of parents' education on reading achievement are estimated with structural equation modelling.

Results. The results reveal that the total effect of parents' education is substantial and that almost half of this effect is mediated through other variables, i.e. the number of books at home, early literacy activities, and emergent literacy abilities at the time for school start. The article thus identifies some of the mechanisms through which parents' education exert an influence on children's literacy development.

Conclusions. Cultural reproduction starts in the very early childhood, in informal settings where reading aloud is an important activity. The knowledge of written language that children have at the time for school-start influences further reading acquisition.

Social class achievement differences in preschool and early elementary school is since long known (e.g. Adams, 1990; Coleman *et al.*, 1966; Hanushek, 1986, 1989; Härnqvist, 1992; Snow, Burns, & Griffin, 1998; SOU, 1993; Yang, 2003). The influence on school achievement of a less favourable background in this respect is, however, likely to vary between school systems (OECD, 2001, 2004; SOU, 1993; Yang, 2003). Although the

**Correspondence should be addressed to Eva Myrberg, Department of Education, Gothenburg University, PO Box 300, SE 405 30 Gothenburg, Sweden (e-mail: eva.myrberg@ped.gu.se).*

DOI:10.1348/000709909X453031

socio-economic differences in reading performance between classes and schools in several studies have been somewhat less pronounced in Sweden than in other countries, it is still substantial (e.g. Gustafson, 1998; Myrberg & Rosén, 2006; OECD, 2001; Yang, 2003). Socio-economic status (SES) is a complex and multidimensional concept. The term cultural capital has been used to label the most important dimension of socio-economic influence on school achievement. In most countries, parents' formal educational level has been identified as the key component of cultural capital (Yang, 2003). The aim here is to unfold the influence of home background and more specifically, to reveal some important mediating factors between the educational levels of parents and the reading achievement levels of children.

Reading literacy as a culturally dependent process

Learning to read is related to children's phonological awareness, their orthographic processing skills, their knowledge of letters, their knowledge of the functions of print, and their language skills (Adams, 1990; Cunningham & Stanovich, 1993). The development of literacy is not only a cognitive process though, but also a social process embedded in the relationships in the family and neighbourhood as well as it is a cultural process connected to different practices and preferences among social classes. The simple view on literacy as a phenomenon with altogether positive effects has been questioned. For example, Olson (1994) argued that literacy through history has been used as a means for establishing social control, for turning people into good citizens, productive workers and obedient soldiers rather than for liberation and emancipation (see also Venezky, 1984). Nevertheless, literacy is today not only fundamental for the well-being of individuals but also for democracy development in society.

Bourdieu (1990, 2002; Bourdieu & Passeron, 1977) described how preferences, attitudes, and behaviours that are dominant in higher classes are rewarded by the school system and thus produce social inequalities. The working mechanism for cultural reproduction is by Bourdieu labelled habitus, a system of shared dispositions that generates perceptions, appreciations, and actions such as literacy practices. Habitus is generated by experience but it is also a generating principle that shapes further experience.

The role of parental education

In a previous study of reading literacy levels among students in public and independent schools Myrberg and Rosén (2006) showed that parents' educational level shows a strong relationship with third graders reading achievement regardless of school form. This calls for the further investigation that is made in the present study.

Habitus is explained as cultural capital in an embodied state. Books, dictionaries, etc. on the other hand, are considered as cultural capital in an objectified state and academic qualification is cultural capital in an institutionalized state (Bourdieu, 2002). Bourdieu (1990) described a school diploma for example, as a piece of universally recognized and guaranteed symbolic capital valid not only in an educational field. The concepts of habitus and capital are closely related to the concept of field and none of them can be understood independently (Bourdieu & Wacquant, 1992). What is defined as cultural capital depends on the social field in which the agents operate and the value of a species of capital hinges on the existence of a field in which this competency can be employed.

A basic thesis in the current study is that school achievement partly is an expression of class reproduction. Literacy practices are mediating factors that serve to simplify this reproduction in a stratified society. Literacy, or rather literacy in a variety of shapes, including the utilization of a great number of signs and features in texts, is in literate societies incorporated in the bodies of individuals. As other expressions used in communication and interaction, literacy serves to define a person, the individual appreciations, and practices, as well as the group-membership. The concept of cultural capital as described by Bourdieu (1984, 1990, 2002), is a theoretical hypothesis and a conceptual tool that makes it possible to explain the unequal scholastic achievement of children originating from different social classes by relating academic success to the distribution of cultural capital between classes and class fractions. Literacy is hence conceptualized as a social and political practice rather than a set of neutral skills. Literacy practices can be considered as parts of ones cultural capital. For example, pronunciation or familiarity with certain writers can be parts of ones cultural capital and have distinctive values that will be unrecognized as capital and instead recognized as legitimate competences. In this process, the social determinants of the educational attainment are misrecognized and educational performance is given the value of natural right that distinguishes the dominant group from dominated groups.

The transmission of cultural capital is a long-lasting process that starts in the early childhood. The process of appropriating cultural capital and the time necessary for it to take place mainly depends on the cultural capital embodied in the whole family and the accumulation period covers the whole period of socialization. This transmission of cultural capital is the best-hidden form of hereditary transmission of capital according to Bourdieu (2002). However, as De Graaf, de Graaf, and Kraaykamp (2000) stated, one may argue that cultural capital is a container concept and that it sometimes remains vague how the transmission of cultural codes furthers a child's educational career. The following section brings forward some home activities that in previous research have been identified to relate to children's reading achievement.

Activities in the home that affect reading literacy

Inheritance is most certainly a powerful predictor of individual differences in early reading literacy skills. General verbal intelligence and phonological sensitivity are likely to be key factors (Bowey, 1995). Yet, large environmentally IQ gains between generations suggest an important role for environment in shaping IQ. Dickens and Flynn (2001) argue that even when incorporating very high estimates of heritability there are still strong effects of environment. In addition, there is a strong reciprocal causation between genes and environment that has a multiplier effect where higher IQ leads one into a better environment causing still higher IQ and so on. Hecht, Burgess, Torgesen, Wagner, and Rashotte (2000) found social class differences in growth of decoding and reading comprehension skills every year in a longitudinal study of students from kindergarten to fourth grade. They concluded that very early attainment of reading-related skills substantially influenced individual differences in later reading ability. Noble, Farah, and McCandliss (2006) suggested that SES systematically modulates the association between phonological awareness and reading ability in complex synergizing relations.

Aspects of reading literacy likely to be influenced by the family and home environment include print awareness, knowledge of narrative structure, literacy as a source of enjoyment, and vocabulary and discourse patterns. Snow *et al.* (1998) listed in an overview some critical conditions that in addition to each child's intellectual and sensory capacities, appear to contribute to successful reading by schoolchildren. These are above all: positive expectations about and experiences with literacy from an early age, support for reading-related activities, and instructional environments conducive to learning. In practice, these areas may be highly interrelated. They are also likely to vary with social class.

The value placed on literacy in a family may at least partly be expressed by reading material in the home. The number of books at home is a variable well known to be associated with students' achievement and it has been used in research as an indicator of children's availability of literacy resources (see for example Elley, 1992). The mere availability of print and frequent exposure to texts of different kinds may have benefits for young children's literacy and may promote the development of specific skills and knowledge that enables more efficient subsequent reading. Furthermore, environments that are rich in literature offer many opportunities for joint activities between parents and children (Paris & Cunningham, 1996).

Moschovaki (1999) described literary experiences as to a great extent embedded in family social events rather than being pure reading and writing in itself. Initially, children become aware of the existence of print in the process of doing something as part of their everyday experiences, when shopping, walking in the street, or watching television. Inside the family context, education occurs at every moment whether it is deliberate and systematic or is realized in an informal way through everyday activities. A set of attitudes, expectations, and information is transmitted and this constitutes informal instruction. Snow *et al.* (1998) put forward the greater opportunities for children from middle-income homes than for children from low-income homes to informal literacy learning like library visits or play with print. Paris and Cunningham (1996) pointed out that some preschoolchildren have as many as 1,000 h of shared reading with parents while others may begin kindergarten without much experience of, or adult guidance in reading.

Sénéchal and LeFevre (2002) found in a longitudinal study that the link between parents' report of teaching about reading and reading storybooks with children was indirect and mediated through children's emergent literacy skills. The variables that were directly related to reading skills at the end of grade 1 where those most closely tied to the mechanics of reading. The pathways for reading achievement in grade 3 were different. Reading storybooks at home predicted children's receptive language skills both concurrently and longitudinally and this relation held after controlling for exogenous factors such as parents' education and endogenous factors such as phonological awareness. Sénéchal and LeFevre found that children's exposure to storybooks at home began to show a strong effect on reading performance once the mechanics of reading were under control and children were reading more fluently (see also Jonsson, 1994; Snow *et al.*, 1998).

Long-term effects of early literacy attainment

Existing evidence suggests a continuous influence of individual differences in early reading skills from kindergarten onward (Sénéchal & LeFevre, 2002; Stanovich, 1986) although schooling may have equalizing effects. Lundberg (1985) found children's reading abilities in grade 1 to be good predictors of reading achievement in grade 3. Paris and Cunningham (1996) also concluded from an overview that reading skills in kindergarten and first grade predicted reading skills later in second grade and that there is developmental continuity in school success or failure based on the achievement of children in primary grades. Stanovich (1986, 2000) has put forward the Matthew effect, meaning that individual differences in reading performance tend to increase during the school years where the rich get richer and the poor get poorer. Students with difficulties in reading skills read less and choose easier texts. As reading is fundamental for learning in all subjects this process affects the overall knowledge acquisition over time.

Hecht *et al.* (2000) found in a longitudinal study that very early attainment of reading-related skills can considerably influence individual differences in later reading ability, even after several years of formal instruction in elementary school. Social class differences in growth of decoding and reading comprehension were in this study found during each time interval that was considered (kindergarten, first grade, second grade, third grade, and fourth grade) while controlling for prior decoding skills. The greatest attenuation of SES effects in growth of reading skills occurred when kindergarten levels of print knowledge were accounted for. Hecht *et al.* suggested early levels of print knowledge to be the most important mediators of social class differences in growth of decoding and reading comprehension skills through children's fourth grade year.

Ever since 1966 when Coleman *et al.* reported that the educational deficit of children from low-income families was present at school entry and increased with every year they stayed in school, evidence of SES differences in reading performance has continued to accumulate (Hanushek, 1989; Snow *et al.*, 1998). Also, though there is vast evidence of the influence of cultural capital on students' achievement, the relative effects of various mediating factors are not yet fully investigated. The present analysis aims to estimate the total effect of parents' education on children's reading achievement, to identify mediating factors and estimate their relative effects. Institutionalized cultural capital in the form of parents' education is hypothesized to be mediated through the number of books at home, early reading activities with the children during the preschool years and the children's early reading abilities. The study investigates the relative importance of these explaining factors.

Method

The following sections present the design of the study including sample and variable descriptions. Methods and analysis procedures are also described.

Data source – the IEA PIRLS 2001 study, design, and samples

The study relies on data from Progress in International Reading Literacy Study (PIRLS) 2001 conducted by the The International Association for the Evaluation of Educational Achievement (IEA; Mullis *et al.*, 2003). The comparative study PIRLS is designed to study reading literacy and related factors among 9- to 10-year-olds every 5th year. The first data collection took place in 2001 with 35 countries participating. Sweden participated with three separate samples of students of students. The design of the international study is described in detail in the PIRLS framework (Campbell *et al.*, 2001) as well as in the *PIRLS 2001 technical report* (Martin *et al.*, 2003). The analyses in the present study are based on two representative grade 3 samples, one for PIRLS and one for the repeat of the Reading Literacy study that was administered in 1991, here called RL 2001 (Rosén, Myrberg, & Gustafsson, 2005). In this repeat study the same reading tests as were developed in 1991 were readministered in 2001, while the same parental questionnaire was used in both studies. A total of 10,632 students in 717 classes and 292 schools constitute the sample. About 50% of the students in this study have been administered the tests developed for the PIRLS 2001 study plus one of two test-booklets developed for the Reading Literacy study 1991 (RL, 1991). The other half of the students have been administered both the test-booklets from 1991. The latter booklets thus form a bridge between the samples used in the present study. The instruments used in this study are the reading tests and the home questionnaire responded to by parents. The reading tests were pen and pencil tests with multiple choice questions and open questions. The proportion of students who were absent at the time of testing or who did not receive parental permission to participate in the study was less than 5%.

Variables and descriptives

The variables used in this study and indicating cultural capital in students' families, origin from the home questionnaire. Parents' educational level is used as independent variable. It is measured on a scale with eight levels: from not finished compulsory school to Bachelors or Masters exam. Indicators of number of books at home, early reading activities with the child and the child's early reading abilities are used as mediating variables.

A total score of reading achievement is used as outcome variable. The scales for RL 2001 and PIRLS 2001 were developed in accordance with item response theory (IRT-scaling). In short, IRT scaling methodology produces a score on the basis of the responses of each student to the items that he or she took, taking into account the difficulty and discriminating power of each item. The model describes the probability that the student will answer in a specific way to an item in terms of the respondent's proficiency, which is an unobserved or latent trait. The result is an interval scale (Gonzales, 2003). There are two different reading scales, one for PIRLS and one for the repeat of the RL-study 1991. The two scores have in this study been transformed to the same scale by equating the percentile scores of the two scales. That is, the 10% of the students who achieved the highest scores in the PIRLS-test were assigned the same test scores as the 10% highest achieving students who took the RL-test and so on (Skarlind, 2005). The appropriateness of such equating is supported by another study where a multivariate analysis of the dimensionality of the two reading achievement tests showed that the general reading achievement dimensions correlate .99 between the two tests (Gustafsson & Rosén, 2006). In order to further validate the results, separate analyses of the two samples have been made using two different outcome variables, one is the mean of five plausible values for the RL-study and the other one is the mean of five plausible values for PIRLS. The two international total achievement scores have a mean of 500 and a standard deviation of 100. The mean of the Swedish third grade students on the total score of reading achievement used here is 522 points. Selected variables are described in Table 1.

Descriptives for the manifest variables including means, standard deviations, and number of cases are shown in Table 2.

Included in the sample used in this study are the students for whom a total achievement score has been estimated. For most of the variables from the home questionnaire the percent missing is around 10% though not surprisingly, somewhat higher for fathers' education, probably depending on a number of single mother households.

Covariances and correlations among the manifest variables have been estimated and are presented in Table 3.

Table 1. Indicators of reading achievement and social background of students

Variables	Information/question/statement	Source
Parents' educational level		
Mothers' education	Mother's educational level Eight alternatives: some compulsory school, completed compulsory school, 2 years of upper secondary education, 3 years of upper secondary education, post secondary education, up to 2 years of university studies, up to 3 years of university studies, Bachelors or Masters exam	Parent
Fathers' education	Father's educational level Eight alternatives: some compulsory school, completed compulsory school, 2 years of upper secondary education, 3 years of upper secondary education, post secondary education, up to 2 years of university studies, up to 3 years of university studies, Bachelors or Masters exam	Parent
Number of books at home		
Books at home	About how many books are there in your home? Five alternatives: 0–10, 11–25, 26–100,101–200, more than 200	Parent
Children's books	About how many children's books are there in your home? Five alternatives: 0–10, 11–25, 26–50, 51–100, more than 100	Parent
Early reading activities		
Read with child	Before your child began grade 1, how often did you read books with him or her? Three alternatives: never/almost never, sometimes, often	Parent
Stories to child	Before your child began grade 1, how often did you tell stories to him or her? Three alternatives: never/almost never, sometimes, often	Parent
Early reading abilities		
Rec letters	How well could your child recognise most of the letters in the alphabet when he/she began grade 1? Four alternatives: not at all, not very well, moderately well, very well	Parent
Read words	How well could your child read some word when he/she began grade1? Four alternatives: not at all, not very well, moderately well, very well	Parent
Read sentences	How well could your child read some sentences when he/she began grade 1? Four alternatives: not at all, not very well, moderately well, very well	Parent
Students' reading achievement		
Reading achievement	Total achievement on PIRLS 2001 or RL 1991/2001 reading achievement tests (IRT-scores)	Student

All correlations were significant at the 1%-level. Indicators of early reading abilities showed the highest correlations with students' achievement score. It should be noted though, that the correlations are affected by measurement error since the estimations derive from observed variables.

Table 2. Means, standard deviations, and number of cases for manifest variables

	Mean	*SD*	*N*
Mothers' education	4.84	2.06	9,402
Fathers' education	4.51	2.16	9,087
Books at home	3.93	1.11	9,579
Children's books	3.86	1.04	9,574
Read with child	2.69	0.50	9,562
Stories to child	2.26	0.66	9,514
Rec letters	3.23	0.76	9,510
Read words	2.77	0.98	9,474
Read sentences	2.07	1.01	9,482
Reading achievement	522.76	78.97	10,632

Methods of analysis

The major aim of the present study is to explore the influence of parents' education on reading achievement, including factors that mediate this influence. The operationalization of the conceptual model relies heavily on the three aspects of cultural capital described in the introduction that has been formulated by Bourdieu (2002), that is, institutionalized capital in the form of school diplomas, objectified capital in the form of home library and (to less extent in this study) embodied capital in the form of personal dispositions. Structural equation modelling (SEM) technique with latent variables is the main method of analysis.

Latent variables reflect hypothetical constructs and are estimated from the relations among two or more indicators. This is done with confirmatory factor analysis which estimates and tests a hypothesized model against the observed covariances among observed variables.

Variability of an indicator is composed of two major components, systematic, and non-systematic variance, the latter also referred to as random error. Systematic variance may be due to various sources; the latent variable that the indicator is presumed to affect, other latent variables, the measurement method used, and systematic errors as for example social desirability. The latent variables have the advantage of being free from non-systematic variance which otherwise causes regression coefficients to be underestimated (Pedhazur & Pedhazur-Schmelkin, 1991).

In a first step of the analysis, a measurement model was set up, defining four latent variables. The purpose of measurement models is to improve the reliability of intended measures and thereby improve reliability. In a next step, a regression model was fitted to data. In this model, the latent variables were assumed to be correlated and each of the four latent variables was assumed to have direct effects on reading achievement. The model presumes that the latent variables just are correlated, there being no ordering of the variables in terms of independent and dependent. It is however, possible to impose a hypothesized structure of relationships among the latent variables in a so-called structural relations model or path model with latent variables (Bollen, 1989; Jöreskog & Sörbom, 1993; Loehlin, 2004). This was done in the third step where the conceptual model employed theoretically satisfies an important criterion in research with explaining ambitions (see Aneshensel, 2002). It follows a chronological order where parents' education precedes the number of books at home, the reading activities with the preschool child, the child's emergent literacy in the beginning of first grade and the reading achievement score in grade three.

Table 3. Covariances and correlations among manifest variables

	Reading achievement	Mothers' education	Fathers' education	Books at home	Children's books	Read with child	Stories to child	Rec letters	Read words	Read sentences
Reading achievement	6,235.94	37.68	38.10	24.54	20.32	8.60	2.77	22.41	28.45	26.25
Mothers' education	0.24	4.25	2.20	0.85	0.58	0.23	0.13	0.18	0.17	0.12
Fathers' education	0.23	0.50	4.65	0.77	0.45	0.18	0.14	0.21	0.23	0.19
Books at home	0.28	0.38	0.33	1.23	0.70	0.17	0.09	0.12	0.11	0.04
Children's books	0.25	0.27	0.20	0.61	1.07	0.19	0.09	0.13	0.11	0.43
Read with child	0.22	0.22	0.16	0.31	0.36	0.25	0.09	0.07	0.07	0.04
Stories to child	0.05	0.09	0.10	0.12	0.13	0.27	0.43	0.03	0.05	0.06
Rec letters	0.38	0.12	0.13	0.14	0.16	0.18	0.07	0.57	0.52	0.46
Read words	0.38	0.09	0.11	0.10	0.11	0.14	0.08	0.72	0.92	0.76
Read sentences	0.33	0.06	0.09	0.04	0.04	0.08	0.08	0.60	0.79	1.03

Note. All correlations significant at the 1% level. Covariances in and above the diagonal, correlations below the diagonal.

Model fit was evaluated with the chi-square test, which tests whether the model's implied covariance matrix agrees with the observed covariance matrix. However, because the large sample size of the current study causes the chi-square statistics to become large even with small deviation between model and data, more reliance was put on the RMSEA and CFI indices which measure the amount of discrepancy between model and data.

Case weights, the 'complex sample' option and the 'missing data' option in Mplus (Muthén & Muthén, 2001) were used in order to account for stratification, cluster effects, and cases with incomplete data. Mplus was used under the modelling environment of STREAMS (Gustafsson & Stahl, 2001).

Results

In the following, the results of the three major steps of the analysis are presented, a measurement model, a regression model, and finally a structural equation model. In addition, the structural model has been estimated with a split sample. The measurement model in Figure 1. Defines four latent variables that are reflected in two or more observed indicators, respectively. The latent variable *Parental education* is measured by the two observed variables 'Fathers' education' and 'Mothers' education'. *Books at home* derives from 'Books at home' and 'Children's books'. *Early reading activities* is constructed out of 'Read with child' and 'Stories to child'. Finally, *Early reading abilities* derives from 'Rec Letters', 'Read words', and 'Read sentences'. Reading achievement is measured by a single observed variable, 'Reading achievement'.

In this model, the latent variables are independent variables which influence the observed variables. This is illustrated by the unidirectional arrows. The latent variables are in the measurement model assumed to be freely correlated which is shown by the curved lines.

The relationship between manifest and latent variables are expressed by standardized loadings, which conceptually can be thought of as standardized regression coefficients. Hence, by squaring the standardized loadings, the amount of variance in the observed variables accounted for by the variables is obtained. About 60% of the variance in the latent variables is accounted for by the observed variables. The improvement in reliability using latent variables instead of observed variables is thus substantial. As the latent variables are free from measurement errors the strong relations between observed and latent variables indicate that the latent constructs are fairly well measured.

According to the chi-square test the model fit was not so good ($\chi^2 = 471.8$, $df = 22$, $p < .00$). However, the statistical test was heavily influenced by the very large sample size, and the two most commonly used descriptive indices of model fit indicated a good fit between model and data, CFI = .98 and RMSEA = .04.

The next model is a regression model where the latent variables are assumed to be correlated and all latent variables have direct effects on the observed reading achievement variable (Table 4).

Again the chi-square test did not indicate especially good fit, $\chi^2 = 541.3$, $df = 26$, $p < .00$. However, the other indices used, indicated a good fit between model and data (CFI = .98, RMSEA = .04). All relations were significant at the 1%-level. *Parental education* and *Books at home* showed a high correlation (.58) and so did *Books at home* and *Early reading activities* (.50). There was also a substantial correlation between *Early reading abilities* and 'Reading achievement' (.35). The effect of *Parental*

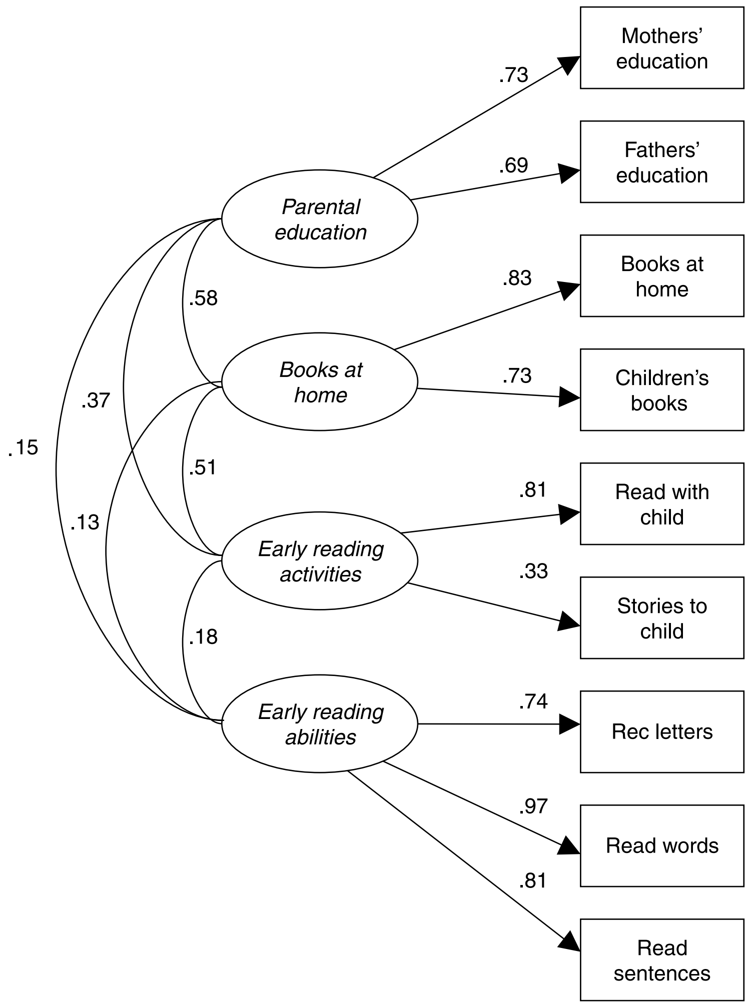

Figure 1. Measurement model with observed and latent variables indicating cultural capital in families.

education on 'Reading achievement', was estimated to .17, the effect of *Books at home* was .17 and the effect of *Early reading activities* was .06.

In the next model the relation between the four latent variables were specified and tested against the data. This model is an attempt to explicate and elaborate the mediating factors and mechanisms that explain why there is a relationship between *Parental education* and 'Reading achievement'. In the language of Aneshensel (2002) this is the focal relationship and through involving a set of mediating variables, the relation between *Parental education* and 'Reading achievement' is at least partly explained. Within the SEM framework, this is most easily done by decomposing the total effect of *Parental education* on 'Reading achievement' into a set of direct and indirect effects.

Parental education is the only independent variable in the final model presented in Figure 2. Its causal sources lie external to the path diagram. In a path diagram the correlation between any two variables can be expressed as the sum of the compound paths connecting these two points. The numerical value of a compound path is equal to the product of the values of its constituent arrows. When a correlation between two

Table 4. Regression model with correlations among latent variables and direct effects on reading achievement

	Parental education	Books at home	Early reading activities	Early reading abilities
Parental education				
Books at home	.58			
Early reading activities	.15	.50		
Early reading abilities	.15	.14	.18	
Reading achievement	.17	.17	.06	.35

variables is due to the presence of one or more additional variables we consider the relationship as indirect (Loehlin, 2004).

The hypotheses formulated in the introduction about patterns of influence among variables representing cultural capital, literacy practices at home, and early reading abilities imply that we expect the following relationships: *Parental education* affects *Books at home*, *Early reading activities, Early reading abilities* and Reading achievement. *Books at home* affects *Early reading activities*, *Early reading abilities* and 'Reading achievement'. *Early reading activities* affects *Early reading abilities* and 'Reading achievement'. *Early reading abilities* affects 'Reading achievement'. This set of hypothesized relations is tested in the structural model. The *t* value for the effect of *Books at home* on *Early reading abilities* was however non-significant (.65) and this effect was thus removed from the final model presented in Figure 2. Direct, indirect and total effects are also presented in Table 5.

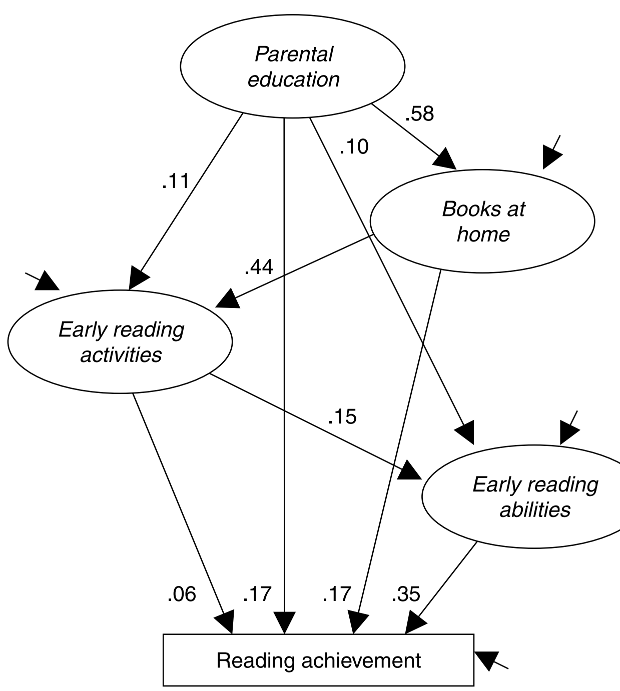

Figure 2. Structural equation model with direct and indirect effects of parents' education on students' achievement.

Table 5. Direct, indirect, and total effects of parental education on reading achievement

	Direct	Indirect	Total
Parental education	.17	.17	.34
Books at home	.17	.05	.22
Early reading activities	.06	.05	.11
Early reading abilities	.35		.35

The fit indices did not differ much between models. In this latter model $\chi^2 = 545.8$, $df = 27, p < .00$, CFI = .98, RMSEA = .04. The model was thus accepted as providing a good representation of the relations among the variables. The similar loadings in the regression model and the structural model also indicate that the model is fairly robust.

The structural model presented as a graph in Figure 2 is to be read vertically from the top to the bottom and the variables are arranged in a chronological and logical order. The starting-point is the latent variable *Parental education*.

While the direct effect of *Parental education* on 'Reading achievement' is modest (.17), the total effect is substantial (.34). This estimate agrees well with what has been found in previous research (SOU, 1993; White, 1982; Yang, 2003). The reason why the total effect is stronger than the direct effect is that there are indirect effects of *Parental education* that are mediated through other variables. The strongest effects go via *Number of books*, of which the most important one is directly from books at home to achievement (.17). There are also two minor indirect effects that are mediated through *Early reading activities*, one that goes directly to 'Reading achievement' and one via *Early reading abilities*. Finally, there is an indirect effect of *Parental education*, which goes via *Early reading abilities*. The model thus explains a part of the effect of parents' education on achievement in terms of books at home and use of those books for literacy purposes.

Books at home has a total effect of .22 on 'Reading achievement' of which the greatest part is a direct effect (.17). The indirect effects are mediated through *Early reading activities* and *Early reading abilities.* The total effect of *Early reading activities* on 'Reading achievement' is .11 of which .06 is a direct effect and .05 is mediated via *Early reading abilities*. The result shows that the number of books of home is exerting an important influence over students' reading achievement in an indirect way by informal shared reading and emergent literacy but it also has a direct relationship with students' achievement.

The last explanatory variable in the chain is *Early reading abilities* which has a total effect of .35 on 'Reading achievement'. This effect shows that the retrospective assessments made by the parents of the students' reading competence when starting school correlate quite highly with measured reading achievement in grade 3. The effect also supports earlier research previously presented in the introduction, suggesting emergent literacy skills to affect later reading achievement levels.

Discussion and conclusions

Parents' educational level, or in the words of Bourdieu, their institutionalized cultural capital, has in this study been shown to exert an important influence on students' reading performance. Thereby, children's reading achievement levels can be described partly as manifestations of the reproduction of cultural capital within families.

Taking a closer look at the estimated loadings in the path model, several interesting findings appear. There is a direct effect of parents' educational level that neither is explained by the literary environment, that is, the number of books at home, nor is connected to literary activities at home. The estimated direct effect of parents' educational level on students' reading achievement may be an indicator of educational expectations and aspirations for the child. Theories by Bourdieu regarding cultural reproduction support such an operative mechanism.

Books at home is an important mediator of parents' educational level and in the language of Bourdieu, an expression of objectified cultural capital in the homes. A correlation between the number of books at home and school achievement is a common finding in research. However, from this study we can conclude that early reading activities with children in the home mediate a great part of the influence of books. The total effect of books at home as estimated in earlier studies (e.g. Elley, 1992; Taube, 1995) can be described as to large extent unexplained as long as one does not have any indicators of reading activities with children. Early reading activities in turn, affect early reading abilities. The estimated amount of direct influence of the number of books at home on achievement may reflect the parents' own reading interest, and the value parents place on reading or on the world of print. It might also be so that the number of books expresses an emphasis on cultural issues and refinement. However, as the effect of the number of books at home on children's early reading abilities was non-significant, it can be tentatively suggested that the mere existence of books in homes not is enough to inspire children's reading.

The direct effect of early reading abilities on reading achievement in the third grade was quite large. This result agrees with existing evidence that suggests early individual differences in literacy skills to exert an influence at least throughout the first years of schooling.

In addition, a sizeable effect runs from parents' education via books at home, through early reading activities, early reading abilities, and further to students' achievement. The strong relationship to 'read with child' suggests that the essence of the latent variable *early reading activities* is reading aloud. As was described in the introduction, previous research has shown that shared enjoyment with words and texts is not necessarily related to teaching print *per se*. High educated parents are more likely to engage children in literacy activities in their everyday life. In shared book reading the children may learn to pay attention to features of the written text, they might be asked to predict events in stories and being encouraged to ask questions about form and content. In doing so, a set of values, attitudes, expectations, and information can be transmitted, that alludes to the elaborated code described by Bernstein (1971). This in turn, makes children perceive themselves as readers and writers long before they can actually read and write (see for example, Paris & Cunningham, 1996 or Snow *et al.*, 1998). Following Bourdieu, the children's habitus have been shaped in a direction that facilitates further reading acquisition.

The effect of parents' education on third graders reading achievement scores is thus somewhat more unfolded in the path analysis. On average, well-educated parents not only have more books at home than have the less educated, they also use their knowledge of books and written language to create an educational environment for their children where reading aloud is an important activity. This influences the emergent literacy of first graders but it also continues to exert an influence over the reading achievement of third graders.

It is well known that retrospective statements have disadvantages in the form of bias and that there is a tendency to judge whatever past phenomena in the light of present knowledge. This may be a limitation of this study that, unfortunately, cannot be overcome with available cross-sectional data. Another limitation with this study is that there are no possibilities to take individual differences like, for example, language skills into account. However, theoretically it is not possible to find a specific age or point where SES starts to exert influence. According to for example Dickens and Flynn (2001), Hecht *et al.* (2000), and Noble *et al.* (2006) one can expect genes and family environment to have a reciprocal relationship from a very early age. There is, however, still much to be done in this area of research.

Acknowledgements

The study has been financed by the Swedish National Agency of Education and the Swedish Research Council.

References

Adams, J. M. (1990). *Beginning to read. Thinking and learning about print*. Cambridge, MA: The MIT Press.

Aneshensel, C. S. (2002). *Theory-based data analysis for the social sciences*. London: Sage.

Bernstein, B. (1971). *Class, codes and control. Vol. 1. Theoretical studies towards a sociology of language*. London: Routledge and Kegan Paul.

Bollen, A. K. (1989). *Structural equations with latent variables*. New York: Wiley.

Bourdieu, P. (1984). *Distinction: A social critique on the judgement of taste*. Cambridge, MA: Harvard University Press.

Bourdieu, P. (1990). *In other words. Essays towards a reflexive sociology*. Cambridge: Polity Press.

Bourdieu, P. (2002). The forms of capital. In A. H. Halsey, H. Lauder, P. Brown, & A. Stuart Wells (Eds.), *Education – culture economy, society* (pp. 46–58). Oxford: Oxford University Press.

Bourdieu, P., & Passeron, J.-C. (1977). *Reproduction in education, society and culture*. Beverly Hill, NJ: Sage.

Bourdieu, P., & Wacquant, L. J. D. (1992). *An invitation to reflexive sociology*. Cambridge: Polity Press.

Bowey, J. A. (1995). Socioeconomic status differences in preschool phonological sensitivity and first-grade reading achievement. *Journal of Educational Psychology*, *87*(3), 476–487.

Campbell, J. R., Kelly, D. L., Mullis, I. V. S., Martin, M. O., & Sainsbury, M. (2001). *Framework and specifications for PIRLS assessment 2001* (2nd ed.). Chestnut Hill, MA: Boston College.

Coleman, J. S., Cambell, E. Q., Hobson, C. J., McPartlant, J., Mood, A. M., Weinfeld, F. D., *et al.* (1966). *Equality of educational opportunity*. Washington, DC: U.S. Government Printing Office.

Cunningham, A. E., & Stanovich, K. E. (1993). Children's literacy environments and early word recognition skills. *Reading and Writing: An Interdisciplinary Journal*, *5*, 193–204.

De Graaf, N. D., de Graaf, P. M., & Kraaykamp, G. (2000). Parental cultural capital and educational attainment in The Netherlands: A refinement of the cultural capital perspective. *Sociology of Education*, *73*(2), 92–111.

Dickens, W. T., & Flynn, J. R. (2001). Heritability estimates versus large environmental effects: The IQ paradox solved. *Psychological Review*, *108*(2), 346–369.

Elley, W. B. (1992). *How in the world do students read?* The Hague: IEA.

Gonzales, E. J. (2003). Scaling the PIRLS reading assessment data. In M. O. Martin, I. V. S. Mullis, & A. M. Kennedy (Eds.), *PIRLS 2001 technical report* (pp. 151–168). Chestnut Hill, MA: Boston College.

Gustafson, J.-E. (1998). Social background and teaching factors as determinants of reading achievement at classroom and individual level. *Nordisk Pedagogik*, *18*(4), 241–250.

Gustafsson, J.-E., & Rosén, M. (2006). The dimensional structure of reading assessment tasks in the IEA reading literacy study 1991 and the progress in international reading literacy study. *Educational Research and Evaluation*, *12*(5), 445–468.

Gustafsson, J.-E., & Stahl, P.-A. (2001). STREAMS user's guide (Version 2.5) for Windows 95/98/NT. Mölndal: Multivariate Ware.

Hanushek, E. A. (1986). The economics of schooling: Production and efficiency in public schools. *Journal of Economic Literature*, *24*(3), 1141–1177.

Hanushek, E. A. (1989). The impact of differential expenditures on school performance. *Educational Researcher*, *18*(4), 45–65.

Hecht, S. A., Burgess, S. R., Torgesen, J. K., Wagner, R. K., & Rashotte, C. A. (2000). Explaining social class differences in growth of reading skills from beginning kindergarten through fourth-grade: The role of phonological awareness, rate of access and print knowledge. *Reading and Writing: An Interdisciplinary Journal*, *12*, 99–127.

Härnqvist, K. (1992). Utbildningsreformer och social selektion. [Educational Reforms and Social Selection.]. In R. Åberg (Ed.), *Social bakgrund, Utbildning, Livschanser. [Social Background, Education, Life Chances]* (pp. 71–88). Stockholm: Carlssons.

Jonsson, J. O. (1994). Förskola – en strategi för jämlikhet. [Pre-school – a strategy for equality; in Swedish]. In R. Erikson & J. O. Jonsson (Eds.), *Sorteringen i skolan. Studier avsnedrekrytering och utbildningens konsekvenser. [Sorting in school. Studies of social selection and consequences of education]* (pp. 13–42). Stockholm: Carlssons.

Jöreskog, K.-G., & Sörbom, D. (1993). *Lisrel 8: Structural equation modelling with the SIMPLS command language*. Chicago, IL: Scientific Software International, Inc.

Loehlin, J. C. (2004). *Latent variable models. An introduction to factor, path and structural analysis* (4th ed.). Mahwah, NJ: Erlbaum.

Lundberg, I. (1985). Longitudinal studies of reading and reading difficulties in Sweden. In G. E. Mackinnon & T. G. Waller (Eds.), *Reading research. Advances in theory and practice* (pp. 65–105). New York: Academic Press.

Martin, M. O., Mullis, I. V. S., & Kennedy, A. M. (Eds.), (2003). *PIRLS 2001 technical report*. Chestnut Hill, MA: Boston College.

Moschovaki, E. (1999). Home background and young children's literacy development. *Early Child Development and Care*, *158*, 11–19.

Mullis, I. V. S., Martin, M. O., Gonzales, E. J., & Kennedy, A. M. (Eds.), (2003). *PIRLS 2001 international report. IEA's study of reading achievement in 35 Contries*. Chestnut Hill, MA: Boston College.

Muthén, L. K., & Muthén, B. O. (2001). *Mplus user's guide*. Los Angeles, CA: Muthén.

Myrberg, E., & Rosén, M. (2006). Reading achievement and social selection into independent schools in Sweden – results from IEA PIRLS 2001. *Scandinavian Journal of Educational Research*, *50*(2), 185–205.

Noble, K. G., Farah, M. J., & McCandliss, B. D. (2006). Socioeconomic background modulates cognition–achievement relationships in reading. *Cognitive Development*, *21*, 349–368.

OECD (2001). *Knowledge and skills for life. First results from the OECD programme for international assessment (PISA) 2000*. Paris: OECD.

OECD (2004). *Learning for tomorrow's world: First results from PISA 2003*. Paris: OECD.

Olson, D. R. (1994). *The world on paper. The conceptual and cognitive implications of writing and reading*. Cambridge: Cambridge University Press.

Paris, S. G., & Cunningham, A. E. (1996). Children becoming students. In D. C. Berliner & R. C. Calfee (Eds.), *Handbook of educational psychology* (pp. 117–147). New York: Macmillan.

Pedhazur, E. J., & Pedhazur-Schmelkin, L. (1991). *Measurement, design and data analysis. An integrated approach*. Hillsdale, NJ: Erlbaum.

Rosén, M., Myrberg, E., & Gustafsson, J.-E. (2005). *Läskompetens i skolår 3 och 4. Nationell rapport från PIRLS 2001 i Sverige*. The IEA Progress in Reading Literacy Study. Göteborg studies in educational sciences 236. Göteborg: Acta Univeritatis Gothoburgensis.

Sénéchal, M., & LeFevre, J.-A. (2002). Parental involvement in the development of children's reading skill: A five-year longitudinal study. *Child Development*, *73*, 445–460.

Skarlind, A. (2005). *Test equating of reading literacy tests in Swedish IEA data from 1970, 1991 and 2001*. Unpublished technical documentation.

Snow, C. E., Burns, M. S., Griffin, P. (Eds.), (1998). *Preventing reading difficulties in young children*. Washington, DC: U.S. National Research Council.

SOU 1993: 85 (1993). In R. Erikson & J. O. Jonsson (Eds.), *Ursprung och utbildning. Social snedrekrytering till högre studier. [Origin and education. Social selection to higher education]*. Stockholm: Department of Education.

Stanovich, K. E. (1986). Matthew effects in reading: Some consequences of individual differences in the acquisition of literacy. *Reading Research Quarterly*, *21*(2), 360–406.

Stanovich, K. E. (2000). *Progress in understanding reading. Scientific foundation and new Frontiers*. New York: The Guilford Press.

Taube, K. (1995). *Hur i all världen läser svenska elever? [How in the World do Swedish Students Read?]*. Stockholm: Skolverket, Rapport 78.

Venezky, R. L. (1984). The history of reading research. In D. P. Pearson (Ed.), *Handbook of reading research* (pp. 3–38). New York: Longman.

White, K. R. (1982). The relation between socio-economic status and academic achievement. *Psychological Bulletin*, *91*(3), 461–481.

Yang, Y. (2003). *Measuring socio-economic status and its effects on individual and collective levels: A cross-country comparison*. Göteborg Studies in Educational Studies, Acta Universitatis Gothoburgensis.

Received 16 February 2006; revised version received 24 April 2009